

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

MEE 416 — Past Questions & Materials
by Renance

Course: MEE 416
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 400

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Paper 1 of 1 — MEE 416 Conditioning and Refrigeration

Level 400 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF ENGINEERING AND TECHNOLOGY

DEPARTMENT OF MECHANICAL ENGINEERING

FIRST SEMESTER EXAMINATION 2024/2025 SESSION

COURSE CODE AND TITLE: MEE 416 AIR CONDITIONING AND REFRIGERATION

TIME ALLOTTED: TWO HOURS, 30 MINUTES (2H:30MINS)

INSTRUCTION: ANSWER ANY FOUR QUESTION

QUESTION 1

- (a) Compare and contrast refrigeration and cooling.
- (b) A refrigerating system operates on the reversed carnot cycle. The higher temperature of the refrigerant in the system is 35°C and the lower temperature is -15°C . The capacity is to be 12 tonnes. Neglect all losses. Determine; vo
- (i) Coefficient of performance :
- (ii) Heat rejected from the system per hour — © na Wroncet ce
- (iii) Power required.

[17.5 Marks]

QUESTION 2

- (a) Describe three (3) methods of producing refrigeration earn fh
- (b) 0.004 kg of water vapour per kg of atmospheric air is removed and temperature of air after removing the vapour becomes 20°C . Determine;
- (i) Relative humidity
- (ii) | Dew point temperature

Assume that condition of atmospheric air is 30°C and 55% relative humidity and pressure is 1.0132 bar

[17.5 Marks]

QUESTION 3

- Sicce(s tce~ (a) Explain the concept of psychrometry and psychrometric
- (b) The air supplied to a room of a building in winter is to be at 17°C and have a relative humidity of 60%. If the barometric pressure is 1.01325 bar, find;
- (i) The specific humidity.
- (ii) The dew point under these conditions.

[17.5 Marks]

QUESTION 4.

- (a) Explain the term coefficient of performance (C.O.P) of a refrigerating system.
- (b) A cold storage plant is required to store 20 tonnes of fish. The temperature of the fish when supplied is 25°C , storage temperature of fish required is -8°C , Specific heat of

fish above freezing point of fish is -3°C . Latent heat of fish $=232\text{kJ/kg}$. If the cooling is achieved within 8hrs, find out; 2 kg /kg vor

(i) Capacity of the refrigerating plant weleu freezing Pitot legs Kg (be

(ii) Carnot cycle C.O.P between temperature range

(iii) If the actual C.O.P is $1/3$ of the carnot C.O.P find out the power required to run the plant. Ne fa

aa ocee {17.5 Marks]

(a) Explain with a neat sketch an air condition cycle.

(b) A refrigeration cycle used freon-12 as the working fluid. The temperature of the refrigerant in the evaporator is -10°C . the condensing temperature is 40°C . The cooling

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